EAS5450: Class #8

Hypothesis-Driven Entrepreneurship

Turning intuitions and enthusiasm into market-driven facts with objective iterative research.

"Devices" ⇒ "Customer-Driven Products"

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"The goal of a startup is to figure out the right thing to build – the thing customers want and will pay for – as quickly as posssible."

Eric Ries
The Lean Startup

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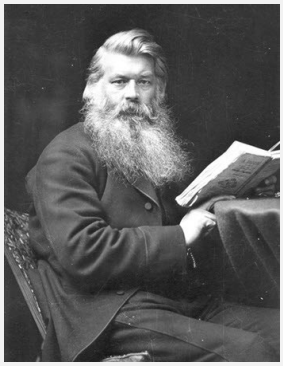
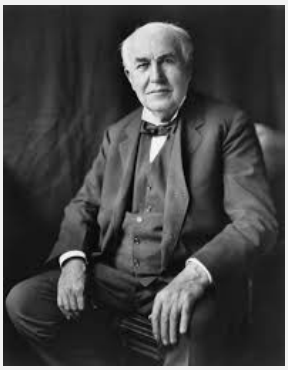
Thomas Edison's invention was market-driven…





Joseph Swan: Inventor of a technology-driven electric lightbulb 1878

- Long life
- Works in parallel
- Compatible with utility grid

Thomas Edison: Inventor of the first ***commercially*** successful lightbulb 1879

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"The real measure of success is the number of experiments that can be crowded into twenty-four hours."

-Thomas Alva Edison

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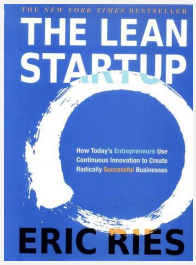
Build-Measure-Learn

De-Risking Through Validated Learning

- Rapid scientific experimentation
  - As quickly as possible
- MVP = "minimum viable product"
- Three P's:
  - Persevere?
  - Pivot?
  - Perish?



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You know "pivot" is overused when...

"I'm not leaving you. I'm pivoting to another man"

Source: *New Yorker*, April 26, 2010

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Get feedback sooner, rather than later…

- Before spending lots of resources, and
- Before accumulating sunk costs,
- Which lead to cognitive biases.

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Recall…

Cognitive Biases

Anchoring on Misleading Perception Impairs Rational Decision-Making

Prof. Daniel Kahneman
Nobel Prize winner

- **Optimism**
  - Mitigate with falsifiable hypotheses
- **Planning Fallacy**
  - Tendency to overestimate benefits
  - Tendency to underestimate costs, durations, risks
- **Confirmation Bias**
  - Over-weighting validating results
  - Ignoring contrary results
- **Sunk Costs**
  - "Wasteful" to discard past effort, cost

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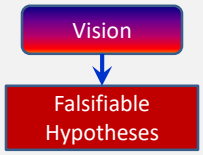

HYPOTHESIS-DRIVEN ENTREPRENEURSHIP

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Falsifiable Hypotheses

- **Can be validated <u>or rejected</u> through decisive experimentation.**
- **If you cannot fail, you cannot learn.**

***BAD**: Our product will spread through word of mouth*

***GOOD**: Our viral coefficient over the next 12 months will exceed 0.5*

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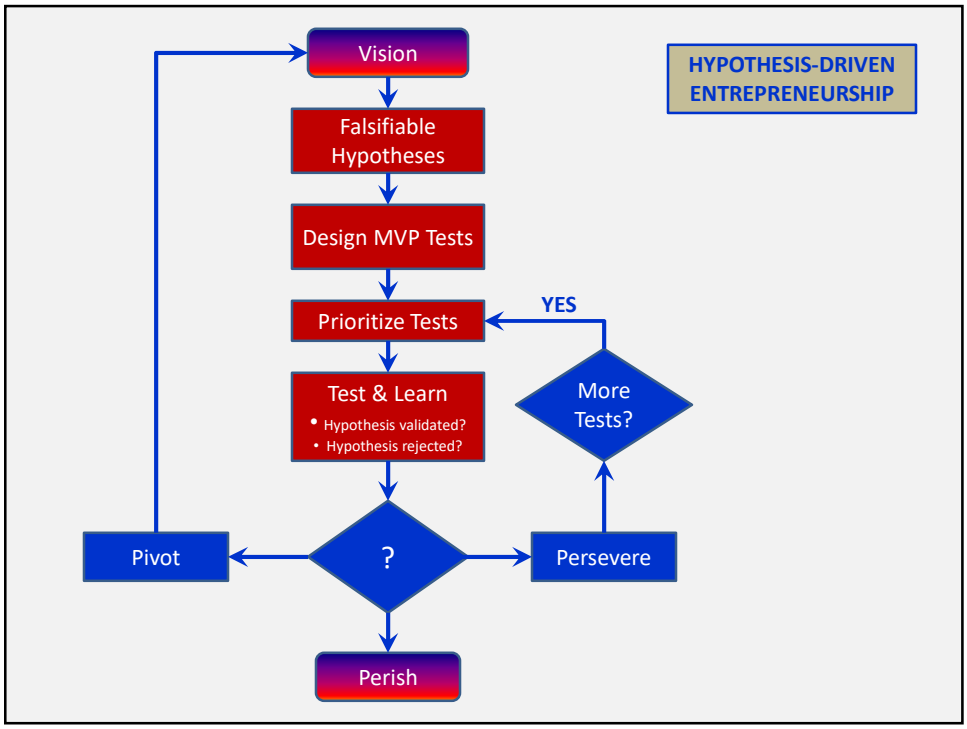

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Recall...

Expect the Unexpected

Recognize False Negatives

"Surprises and reverses can serve as an incentive for great accomplishment. There are no rules here. Just because something doesn't do what you'd planned it to do in the first place doesn't mean it's useless."

"Opportunity is missed by most people because it is dressed in overalls and looks like work."

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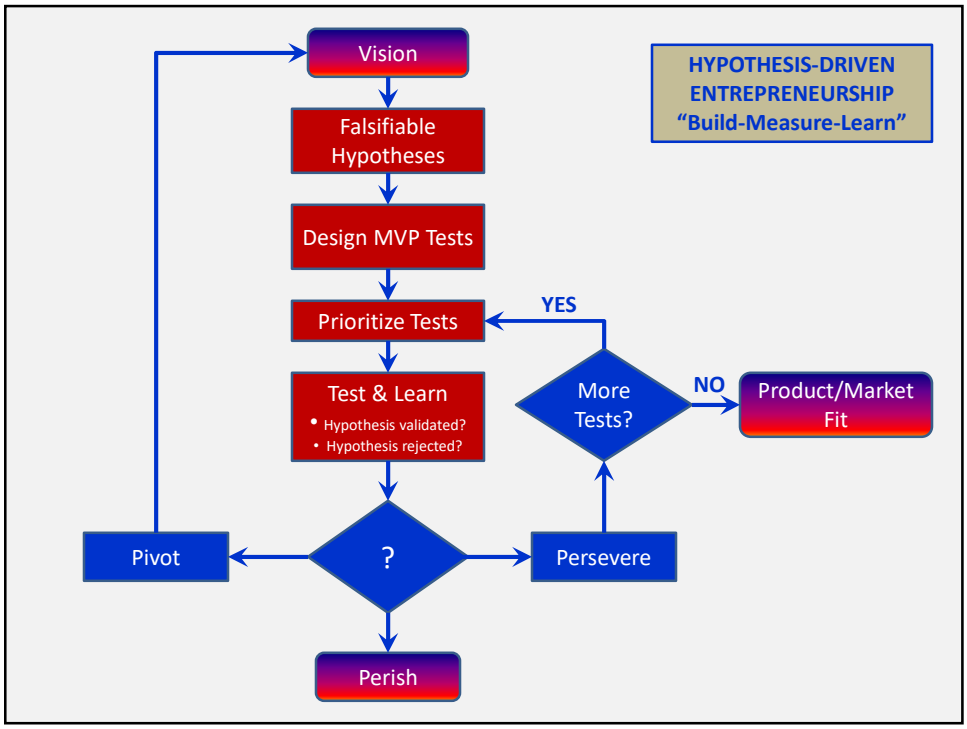

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When MVP Isn't Appropriate…

- Mistakes are not tolerable
- Certainty of demand
- Low quality data
- Long development cycles

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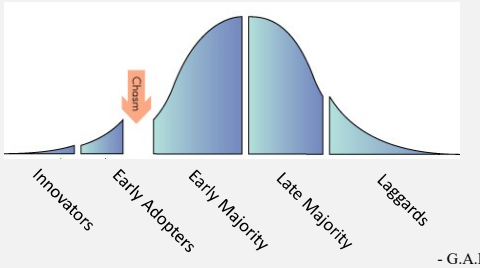
MVP Concerns

- Idea theft
- Reputational risk
- Early adopter vs mainstream psychographics



- G.A.Moore, "Crossing the Chasm"

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A desk is a dangerous place from which to watch the world.

… John le Carré

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Dropbox

"It Just Works"

Agenda:

- Seizing the Opportunity
- Business Model Analysis
- Lean Startup: Hypothesis Testing
- Should Dropbox Target SMBs?
- Epilog

EAS5450 Engineering Entrepreneurship I

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Business Model Analysis

4 Key Elements…

1. Customer value proposition
2. Technology/operations management
3. Go-to-market plan
4. Profit formula

"Business Model Analysis for Entrepreneurs" (HBS No. 812-096)

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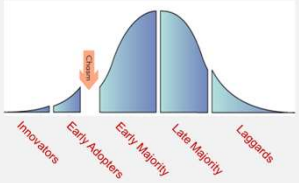
1. Customer Value Proposition

- Customer Problem (Pain)
- Differentiated Solution
- Target Customers
  - Crossing the Chasm
- Pricing



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2. Technology/Operations Management

- Proprietary IP
- In-house vs. Out-source

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3. Go-To-Market Plan

- Viral Marketing
  - Direct network effect
  - Word-of-mouth
  - Casual contact
  - “User-get-user” incentive

- Target Segmentation
  - Avoid corporate gatekeepers
  - Value of targeting individuals

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4. Profit Formula

- Freemium Model
  - Conversion odds over time
- Customer Acquisition Cost (CAC)
- Fixed Costs
- Variable Costs
- Capital Investment
- Working Capital

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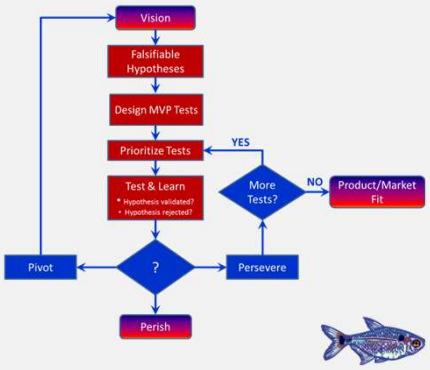
Lean Startup: Hypothesis Testing

- Vision
- Falsifiable Hypotheses
- Minimum Viable Products
  - A/B Testing
- Pivoting
- Product/Market Fit



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Should Dropbox Target SMBs?

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EPILOG

- Jan 2011: $250M Series B led by Index Ventures
- Apr 2013: Launches "Dropbox for Business"
- Jan 2014: $350M Series C led by Blackrock
- Apr 2014: $500M Debt by JB Morgan
- 2014-2016: Exodus from Amazon Cloud
- June 2015: 400 million Dropbox users have created 2.8 billion sharing connections on DropBox and synch 1.2 billion files <u>every</u> <u>day</u>.
- June 2016:
  - 500 million Dropbox users
  - 150,000 enterprise customers
- Sept 2017
  - Market leader: 650 million users
  - SMB & enterprise businesses
  - > $1B/yr revenue
  - Post-money valuation > $10 B
- Mar 2018: $756M Initial Public Offering
- Q2 2024
  - Aug> 700 million registered users
  - 18.2 million paying users
  - $2.5B revenue (TTM): ~11% CAGR (constant currency)

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businesswire

Dropbox Teams Up With McLaren Racing as an Official Technology Partner of McLaren Formula 1 Team

Dropbox is now an official technology partner of McLaren Formula 1 Team to improve organization, collaboration, and ultimately, save precious time, on and off the track

SAN FRANCISCO, February 09, 2024--(BUSINESS WIRE)--Today, Dropbox, Inc. (NASDAQ: DBX) and McLaren Racing announced a multi-year partnership with the McLaren Formula 1 Team for the 2024 season and beyond. The partnership marks Dropbox's first entry into the sport, helping the McLaren team collaborate, share files and content, and save time.

Through the partnership, McLaren will rely on Dropbox for its fast, secure, and reliable syncing and sharing capabilities, in addition to its latest offerings in video collaboration and machine intelligence, to manage vast amounts of content and data, from across the globe.

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IPO March 23, 2018:

- Dropbox IPO priced at $21/share = $756 million
- Opened @ $29/share
- Sept 18, 2024: $24.70/share
- **Feb 10, 2025: $31.98/share**

Yahoo Finance 9/18/2024

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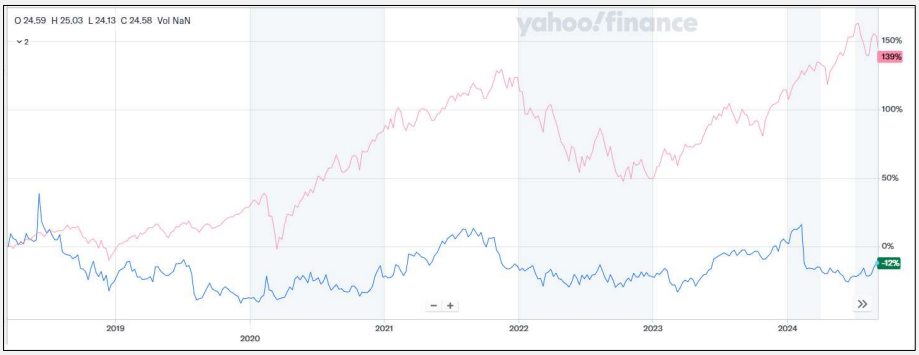
DROPBOX vs NASDAQ

IPO to date



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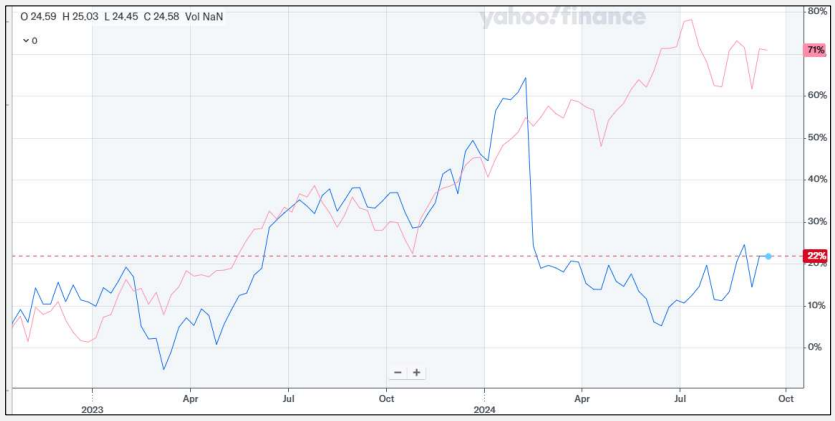
DROPBOX vs NASDAQ

Past 24 Months



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